## Big O Notation

Runtime - the number of computer steps the algorithm takes as a function of the size of the input

Keep only largest term, omit constants

Polynomial → efficient
Exponential → intractable

$O$ → upper bound
$\Omega$ → lower bound
$\theta$ → same rate

## Proving Asymptotic Relations

if $\lim_{n\to\infty} \frac{f(n)}{g(n)} < \infty$, $f(n) = O(g(n))$

if $\lim_{n\to\infty} \frac{f(n)}{g(n)} = c$, $f(n) = \theta(g(n))$ ← $c > 0$

if $\lim_{n\to\infty} \frac{f(n)}{g(n)} > 0$, $f(n) = \Omega(g(n))$

## Recurrence Relations

- function for amount of working needed to solve a problem of size n using smaller subproblems.

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} = O(n^2)$$

$$a + ar + ar^2 + \dots + ar^{n-1} = \frac{a(1-r^n)}{1-r}$$

$$a + ar + ar^2 + \dots = \frac{a}{1-r}$$

for a series $O(1) + O(2) + \dots + O(f(n))$, runtime is $O(f(n))$.

## Divide and Conquer

1. Divide: break large problem into smaller subproblems
2. Conquer: recursively solve each subproblem
3. Combine: combine the answers of the subproblem

## Master Theorem

if relation is of the form $T(n) = aT\left(\frac{n}{b}\right) + O(n^d)$

$$T(n) = \begin{cases} O(n^d) & \text{if } d > \log_b a \\ O(n^d \log n) & \text{if } d = \log_b a \\ O(n^{\log_b a}) & \text{if } d < \log_b a \end{cases}$$

## Graphs

- a set of vertices and directed/undirected edges

DFS: graph traversal that visits all nodes down one path before moving on to another path.

Preorder - order by time when node is first visited

Postorder - order by time when node is done exploring (including all child nodes)

Runtime: $O(|V| + |E|)$

## Edge Types

tree edge: leads to a child, part of the DFS tree
forward edge: leads to a non child descendant
back edge: leads to an ancestor
cross edge: leads to a node that is neither a descendant nor an ancestor

## DFS uses

- path between 2 vertices
- find connected components
- if graph has a cycle
- topological sort of graph

## Topological Sort

- linear ordering of vertices such that for every edge $(u, v)$, vertex $u$ comes before vertex $v$.

Works only on directed, acyclic graphs

## Breadth First Search $O(V+E)$

- Graph traversal that visits all nodes level by level.

UNWEIGHTED

## Strongly Connected Components

- Subset of vertices in which there is a path from every vertex to every other vertex (cycle)

how to find? kosaraju's algorithm

1. reverse all edges in graph G to create $G^R$
2. run DFS and keep track of post order values
3. run DFS on G starting with next available vertex with highest post order value

Repeat until all vertices have been visited.

## Dijkstra's Algorithm (weighted +)

- Finds shortest path from a starting vertex to all vertices in graph.

1. Insert all vertices into PQ, sorting vertices in order of distance from source.
   - Initially, source distance is 0, others are $\infty$
2. Repeat until PQ is empty
   - remove closest/smallest vertex from PQ and relax all edges pointing from v.
   - update distance to neighboring vertices u if $dist[v] + weight(u, v) < dist[u]$

Runtime: inserting $|V|$ times + deleting min $|V|$ times and updating $|E|$ times.

| | Ins | Del | Upd | Tot |
|---|---|---|---|---|
| Array | $O(1)$ | $O(V)$ | $O(1)$ | $O(V^2+E)$ |
| Bin. Heap | $O(\log V)$ | $O(\log V)$ | $O(\log V)$ | $O((V+E)\log V)$ |

## Greedy Algorithms

- Always choose the locally optimal decision (what looks best at the time).
- Prove using exchange argument: Show that the greedy choice is at least as good as any other choice.

## Exchange Arguments

1. Let G be greedy output and O be optimal output
2. Find first place where G and O disagree. G picks g, O picks o.
3. Modify O to include g by removing o if needed, swapping, etc.
4. Show all constraints still hold and value of modified O ≥ value of original O.
5. Therefore, the new O is optimal also. You can repeat for every differing job between O and G. Eventually, O will be modified into G and be optimal so G is also optimal.

## Dynamic Programming

- Reuse information that has already been computed and store it so we do not need to compute it again

1. Define the subproblem
2. Identify the base case
3. Find the recurrence relation
4. Construct the order to solve the subproblems
   - Bottom up approach: Start from base case and build up to n
   - Top down approach: Start from n and recursively call down to base case
5. Find the final solution
6. Solve for runtime and space complexity
   - Time = O(# subproblems · time for 1 subproblem)
   - Space = O(# subproblems)
7. Prove correctness with induction.

## Graph Example

Given graph $G=(V,E)$ and vertex s, let $p(v)$ = # of shortest paths from s to v, $p(s)=1$. Design algorithm to find $p(v)$ for every vertex, given G is undirected and unweighted.

Use BFS. When visiting edge $u \to v$, if v distance is ∞ set to dist[u]+1, set #paths[v] += #paths[u]. Append all unseen neighbors of u to queue.

## Bellman Ford

- For possible negative edge weights

Init dist[s]=0 and all others to ∞.

Repeat V-1 times: Relax edge (u,v) by setting dist[v] = min(dist[v], dist[u] + w(u,v))
Have to do V-1 passes since initially, only source was correct. After 1 pass, first edges are correct, so continue pattern until V-1 (last) edge.

Runtime: O(VE)

## DP Example 1

Knapsack with cap W and n types of items. Item i has weight $w_i$, value $v_i$. Find max total value.

1. Subproblem
DP(i,w) = max val with items 1 to i with cap W

2. Recurrence Relation

$$DP(i,w) = \max \begin{cases} DP(i-1,w) & \text{don't ever use } i \text{ again} \\ DP(i, w-w_i)+v_i & \text{use item } i \end{cases}$$

3. base case
$DP(0,w)=0 \quad DP(i,0)=0$

4. Runtime
# subproblems = O(nW)
time per subproblem = O(1)
Overall = O(nW)

## DP Example 2

n spots for trees, $x_i$ apples at each spot. If tree at i, cannot tree at i-1 or i+1. Find max # of apples

1. Subproblem
DP(i) = max # apples using spots up to i.

2. Recurrence Relation

$$DP(i) = \max \begin{cases} DP(i-2)+x_i & \text{plant at } i \\ DP(i-1) & \text{don't plant at } i \end{cases}$$

3. Base case
$DP(0) = x_0 \quad DP(1) = \max(x_0, x_1)$

4. Runtime
# Subproblems = O(n)
Time per subproblem = O(1)
Overall = O(n)

## D&C Example

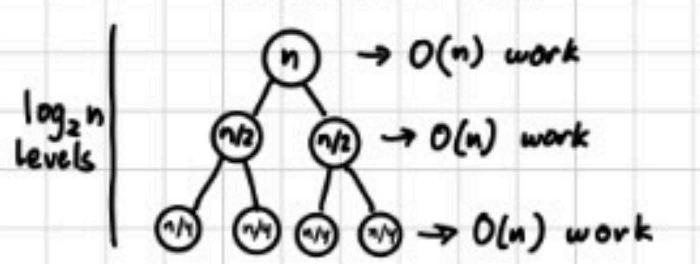
Array $A[0,\dots,n-1]$ where $n>1$ and all pos ints. Find max val of $A[i]+A[j]^2$ s.t. $0 \le i < j < n$

1. Divide: Split array in half
2. Conquer: Recurse to find max val in L and R (ℓ and r)
3. Combine: Find max crossing val
   $z = \max(L) + \max(R)^2$

   return max(ℓ, r, z)

4. Runtime: check max of each half

$T(n) = 2T(\frac{n}{2}) + O(n)$
2 recursive calls




log n levels w O(n) work each
= O(n · log n)

## Backpropagation

Suppose a function $L = L(x_1, x_2, \ldots, x_d)$ that depends on many variables and we want to get all partial derivatives.

Idea: Represent $L$ as a directed, acyclic graph.

Source nodes (in-degree 0) are inputs $(x_1, x_2, \ldots, x_d$ and constants.

Internal nodes store the result of the operations. For example, a node might have $a, b$ as inputs and an addition operation so it stores $c = a+b$ which it passes through its output edge.

Output node (out-degree 0) represents the output $L$.

If there is an edge $u \to v$, $u$ is a child of $v$. The source nodes are leaves, the output node is root.

Forward Pass - Compute the value of each node from the inputs to the output, following topological order.

Cost: $O(|V| + |E|)$, each node is visited once, each edge is traversed once.

Backward Pass - Compute the gradient $\frac{\partial L}{\partial u_i}$ for each node $u_i$ in reverse topological order.

Initialize $\frac{\partial L}{\partial u_n} = 1$, the output node's partial derivative

For $i = n-1, n-2, \ldots, 1$: $\frac{\partial L}{\partial u_i} = \sum_{u_j \in \text{parents}(u_i)} \frac{\partial L}{\partial u_j} \cdot \frac{\partial u_j}{\partial u_i}$.

After the pass, each node stores the correct partial derivative.

Cost: $O(|V| + |E|)$, each node is visited once and each edge is traversed once.

## Fast Fourier Transforms

Idea: Multiply two polynomials together quickly

A degree $d$ polynomial can be determined by its values at $d+1$ points. If you know what the polynomial evaluates to at $d+1$ distinct points, you can find out the polynomial.

For multiplication: If you want $C(x) = A(x) \cdot B(x)$ and you already know $A(x_i)$ and $B(x_i)$ at a bunch of points $x_i$, you can calculate $C(x_i) = A(x_i) \cdot B(x_i)$

1. Pick points $x_0, \ldots, x_{2d}$ that will construct $C(x)$
2. Evaluate $A(x)$ and $B(x)$ at these points
3. Multiply the values to get $C(x_i) = A(x_i) \cdot B(x_i)$
4. Interpolate the $2d$ points to find the coefficients of $C(x)$

Nth Roots of Unity - the $n$ complex numbers that, when raised to the $n$th power, equal 1. They are:

$$w^k = e^{2\pi i k/n} \quad \text{for} \quad k = 0, 1, \ldots, n-1$$

The squares of the $n$th roots are just the $(n/2)$th roots: $(w^k)^2 = e^{2\pi i k \cdot 2/n} = e^{2\pi i k/(n/2)}$

So squaring the $n$ points gives you $n/2$ distinct points.

When you split $A$ into even and odd coefficients:

$$A(x) = A_{even}(x^2) + x \cdot A_{odd}(x^2)$$

$FFT(A(x), n)$ evaluates $A(x)$ at the $n$th roots of unity.

Polynomial Multiplication: $d$ is degree of $A/B(x)$

① Pick the $n$th roots of unity to construct $C(x)$ where $d$ is the degree of the product. Round up to nearest power of two.

② Evaluate $FFT(A(x), n)$ and $FFT(B(x), n)$ to get values at $n$th roots of unity.

③ Multiply corresponding pairs of values to get value representation of $C(x)$

④ Use inverse FFT $= \underset{\text{normalize}}{\frac{1}{n}}\, \underset{\text{same FFT}}{FFT}(C(x) \text{ values}, \underset{w = e^{2\pi i/n}}{w^{-1}})$ to get coefficients.

Runtime: $\underset{\text{Step 2}}{O(n \log n)} + \underset{\text{Step 3}}{O(n)} + \underset{\text{Step 4}}{O(n \log n)} = O(n \log n)$

| Algorithm | Work ($T_1$) | Span ($T_\infty$) | Idea |
|---|---|---|---|
| Parallel Prefix (Scan) | $O(n)$ | $O(\log n)$ | Associativity |
| Carry-Lookahead Add | $O(n)$ | $O(\log n)$ | Linear recurrence → prefix |
| Integer Multiplication | $O(n^2)$ | $O(\log n)$ | 3-for-2 + scan |

## Parallel Algorithms

Model parallel algorithms as directed acyclic graphs.

Nodes - represent a unit of computation

Edges - represent a dependency, which is a constraint that forces one computation to come before another.

Let $T_p$ = time to finish the computation on $p$ processors.

Work $= T_1 = \#$ of vertices in the DAG.
- 1 processor, no parallelism
- $T_p \geq Work/p = T_1/p$, best case scenario.

Span $= T_\infty$ = length of the longest directed path in the DAG (critical path)
- count edges for length.
- cannot do better than $T_\infty$

Speedup $(S_p) = \frac{T_1}{T_p}$. Max speedup is $p$

Efficiency $(E_p) = \frac{T_1}{pT_p}$. Measures how well each processor is utilized. Perfect efficiency = 1, $E_p \leq 1$

Parallelism $= \frac{T_1}{T_\infty}$. High parallelism, processors can stay busy. Low parallelism means the algorithm is mostly sequential.

Brent's Rule - Runtime: $T_p \leq \frac{T_1}{p} + T_\infty$

## 3-for-2 Trick - Carry Save

Three $n$ bit numbers $X, Y, Z$. We want to reduce them to two numbers $S$ and $C$ s.t. $S + C = X + Y + Z$, in constant time on $n$ processors.

Take bit position $k$: add $x_k, y_k, z_k$. sum is between 0 and 3 in format $c_k s_k$.

$s_k = x_k \oplus y_k \oplus z_k$ (XOR - odd number of 1s = 1, even = 0)

$c_k = \text{majority}(x_k, y_k, z_k)$, 1 iff 2 out of 3 are 1s.

$s_k$ is placed at position $k$ of $S$ and $c_k$ is placed at $k+1$ of $C$.

## Integer Multiplication

① Compute partial products (get to the step of needing to sum everything).

② Group the $n$ rows into triples and apply the 3-for-2 trick to each triple simultaneously. Repeat. Each round the # of rows shrinks by a factor of 2/3. After $O(\log n)$ rounds, you have exactly two rows.

③ Add the 2 rows using carry lookahead, which takes $O(\log n)$ time.

Total Span $= O(\log n)$ Total Work $= O(n^2)$

## Parallel Prefix (Scan) - Given $a, b, c, d$: outputs $a, a{+}b, a{+}b{+}c, \ldots$

Takes a sequence and an associative binary operator (addition, multiplication, logical AND/OR, minimum, maximum, GCD, matrix multiplication, etc.)

$$y_k = x_1 \circ x_2 \circ \ldots \circ x_k, \quad k = 1, 2, \ldots, n$$

① Reduce - Pair up adjacent elements and compute $z_k = x_{2k-1} \circ x_{2k}$ for $k = 1, 2, \ldots, n/2$ where $\circ$ is the operator. This takes $O(1)$ span since all pairs are independent, so the computations can happen on different processors.

② Recurse - Compute the prefix scan of $z$ recursively. Let $y'$ denote the result: $y'_k = z_1 \circ z_2 \circ \ldots \circ z_k = x_1 \circ x_2 \ldots x_{2k}$

So
$$y'_1 = z_1 = x_1 + x_2 = y_2$$
$$y'_2 = z_1 + z_2 = x_1 + x_2 + x_3 + x_4 = y_4$$
$$y'_3 = z_1 + z_2 + z_3 = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = y_6$$
$$\vdots$$

So $y'_k$ is the prefix of the original sequence up to position $2k$. So we can get all even indexed outputs for free.

③ Expand - Recover all outputs

Even indices: $y_{2k} = y'_k$

Odd indices: $y_{2k+1} = y'_k \circ x_{2k+1}$

All odd-indexed outputs can be computed at the same time, so $O(1)$ span.

Work $= O(n)$ Span $= O(\log n)$

$T_1(n) = T_1(n/2) + O(n)$ $T_\infty(n) = T_\infty(n/2) + O(1)$

This algorithm does no more work than the sequential algorithm.

## Carry Lookahead Addition

① Compute $g$ and $p$ for each bit position in parallel ($O(1)$ span)

$g_i = a_i$ AND $b_i$ — 1 if both $a_i$ and $b_i$ are 1

$p_i = a_i$ XOR $b_i$ — 1 if $a_i \neq b_i$

② Each bit position gets a 2×2 matrix:

$$M_i = \begin{pmatrix} p_i & g_i \\ 0 & 1 \end{pmatrix}$$

Use parallel prefix to do matrix multiplication and compute
$$M_0 = C_1$$
$$M_1 \cdot M_0 = C_2$$
$$\vdots$$

So you will have all carry bits in $O(\log n)$ span, $O(n)$ work.

③ $s_i = a_i$ XOR $b_i$ XOR $c_i$

1 if odd # of 1's, 0 if even.

## Linear Programming

Constructing a program that maximizes a linear objective function given a set of linear constraints

1) Define your variables
2) Find constraints and objective function

Form: $\max \; c_1 x_1 + c_2 x_2 + \ldots + c_n x_n$ Objective Function

s.t. $Ax \leq b$ Constraints

$x \geq 0$ Always needed

Objective Function must maximize. If you want to minimize, negate the function.

Constraints must have $x \leq$ since you are maximizing. If $\geq$, multiply both sides by $-1$.

Feasible Region - set of points that satisfy all constraints
- feasible: there exists at least one solution that satisfies all constraints
- unfeasible: there is no solution that satisfies all constraints
- unbounded: feasible region is open and extends to infinity in some direction

Dual - combines the constraints to get the upper bound of the objective function.
- Weak duality (always holds): the optimal value of the dual will be an upper bound to the optimal value of the primal.
  - primal objective ≤ dual objective
- Strong duality (if primal is feasible and not unbounded): The dual optimal value equals the primal optimal value.

If primal is maximization, dual is minimization

Dual Form: $\min \; y^T b$

s.t. $y^T A \geq c^T$ or $A^T y \geq c$

$y \geq 0$

## Simplex Algorithm - solves linear program

① Start at a vertex

② For all neighboring vertices:
- If there exists a neighbor value > current value, repeat step 2 with neighbor.
- Else, current vertex is optimal value.

Runtime: $O(2^n)$

## Linear Programming Example

Primal LP: $\max \; x_1 + 6x_2$

s.t. $x_1 \leq 200$,
$x_2 \leq 300$,
$x_1 + x_2 \leq 400$,
$x_1, x_2 \geq 0$

Optimal Solution: $(100, 300) \to 1900$
Solve by Simplex algorithm.

For each constraint, assign a value $y_i$:

$y_1$ $x_1 \leq 200$
$y_2$ $x_2 \leq 300$
$y_3$ $x_1 + x_2 \leq 400$

LHS: Replace the coefficients of $x_1$ and $x_2$. $x_1$ is involved in $y_1$ and $y_3$. $x_2$ is involved in $y_2$ and $y_3$. Therefore:

RHS: $200y_1 + 300y_2 + 400y_3$

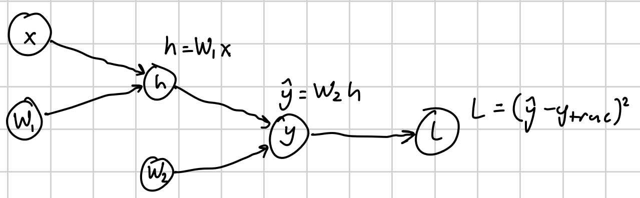
$$(y_1 + y_3)x_1 + (y_2 + y_3)x_2 \leq \underbrace{200y_1 + 300y_2 + 400y_3}_{\text{Dual objective function to minimize.}}$$
$$= x_1 + 6x_2$$

Dual constraints: $y_1 + y_3 \geq 1$
$y_2 + y_3 \geq 6$
$y_1, y_2, y_3 \geq 0$

# Backpropagation Example

$x = \begin{pmatrix}1\\1\end{pmatrix}$ $W_1 = \begin{pmatrix}2 & 1\\0 & 3\end{pmatrix}$ $W_2 = (2\ 1)$ $h = W_1 x$ $\hat{y} = W_2 h$ $y_{true} = 15$

Perform forwards then backpropagation to compute $\frac{\partial L}{\partial x}, \frac{\partial L}{\partial W_1}, \frac{\partial L}{\partial W_2}$ of $L = (\hat{y} - y_{true})^2$




Forward Pass:

$h = W_1 x = \begin{pmatrix}2 & 1\\0 & 3\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}3\\3\end{pmatrix}$

$\hat{y} = W_2 h = (2\ \ 1)\begin{pmatrix}3\\3\end{pmatrix} = 9$

$L = (9-15)^2 = 36$

Back Pass:

$\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y_{true}) = 2(9-15) = -12$

$\frac{\partial L}{\partial W_2} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial W_2} = -12 \cdot (h_1\ h_2) = -12(3\ \ 3) = (-36\ \ -36)$

$\frac{\partial L}{\partial h} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h} = -12 \cdot W_2 = -12(2\ \ 1) = (-24\ \ -12)$

$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial h_1} \cdot \frac{\partial h_1}{\partial x} + \frac{\partial L}{\partial h_2} \cdot \frac{\partial h_2}{\partial x} = -24\begin{pmatrix}2\\1\end{pmatrix} - 12\begin{pmatrix}0\\3\end{pmatrix} = \begin{pmatrix}-48\\-60\end{pmatrix}$

$\frac{\partial L}{\partial W_1} = \frac{\partial L}{\partial h_1} \cdot \frac{\partial h_1}{\partial W_1} + \frac{\partial L}{\partial h_2} \cdot \frac{\partial h_2}{\partial W_1} = -24\begin{pmatrix}1 & 1\\0 & 0\end{pmatrix} - 12\begin{pmatrix}0 & 0\\1 & 1\end{pmatrix} = \begin{pmatrix}-24 & -24\\-12 & -12\end{pmatrix}$

# Parallel Algorithms Example

Given a string of $2n$ parentheses, determine if its valid (every open paren is matched to a later closed paren) in $O(n)$ work and $O(\log n)$ time.

① Build an array where '(' is +1 and ')' is −1. You can do this in parallel.

② Compute all prefix sums in parallel using parallel prefix algorithm. This will compute the running sums up to the current position. If any of the prefix sums < 0, there is an unmatched close paren so it is invalid. The final sum must be 0 so all parenthesis are paired.

Work = $O(n)$ since parallel prefix does $O(n)$ work.

Time = Span = $O(\log n)$ since parallel prefix takes $O(\log n)$ time.

# Parallel Algorithms Example

Given an array of $n$ numbers, remove all negative numbers and output the remaining elements in their original order.

In parallel, you can identify negative elements easily but it is difficult to place each surviving element.

① Build a binary array $P$ where $P[i] = 1$ if $A[i] \geq 0$, else 0.

② Compute prefix sums of $P$ using parallel prefix

③ The prefix sum at $i$ gives the target index $t(i)$ for each element $A[i]$ in the output array.

④ Copy each non-negative $A[i]$ to output$[t(i)]$ in parallel.

Work = $O(n)$ Time = Span = $O(\log n)$

# Linear Programming

Minimize food cost while meeting daily nutritional minimums.

| | | Protein | Carbs | Fat |
|---|---|---|---|---|
| Meat | \$5 | 500 | 0 | 500 |
| Bread | \$2 | 50 | 300 | 25 |
| Shakes | \$4 | 300 | 100 | 200 |

LP: $\min\ 5m + 2b + 4s$ ← min cost: cost per item times amt of item.

s.t. $500m + 50b + 300s \geq 500$
$300b + 100s \geq 100$
$500m + 25b + 200s \geq 400$
$m, b, s \geq 0$

Dual:

$p$ $500m + 50b + 300s \geq 500$
$c$ $300b + 100s \geq 100$
$f$ $500m + 25b + 200s \geq 400$

RHS: $500p + 100c + 400f$

LHS: $m(500p + 500f) + b(50p + 300c + 25f) + s(300p + 100c + 200f)$
$= 5m + 2b + 4s$

Dual: $\max\ 500p + 100c + 400f$

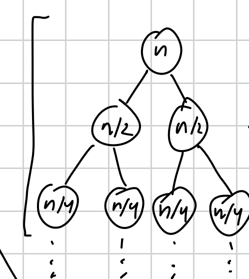
s.t. $500p + 500f \leq 5$
$50p + 300c + 25f \leq 2$
$300p + 100c + 200f \leq 4$
$p, c, f \geq 0$

# FFT Example 1

Let $p(x) = \prod_{i=1}^{n}(x - r_i)$ with distinct roots $r_1, \ldots, r_n$. $p(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_0$. Give an algorithm to compute coefficients $a_0, \ldots, a_n$ and show it runs in $O(n \log^2 n)$.

Given the $n$ roots, split them in two halves. Recurse on each half until you reach the base case where each half has only one factor. Then, start combining the halves using FFT multiplication.

$\log_2 n$ levels




$T(n) = 2T(n/2) + O(n \log n)$

→ Each level does $O(n \log n)$ work

$\log n \cdot O(n \log n) = O(n \log^2 n)$

# FFT Example 2

Given an array $A$ of $n$ integers where each value is between 0 and $n$, determine if any three elements (with repetition allowed) sum to exactly $n$.

$A[i] + A[j] + A[k] = n$ is equivalent to $x^{A[i]} \cdot x^{A[j]} \cdot x^{A[k]} = x^n$

Let $f(x) = x^{A[0]} + x^{A[1]} + \ldots + x^{A[n-1]}$. When you multiply $(f(x))^3$, the coefficient of $x^n$ is the number of ways to pick three monomials from $f(x)$ whose exponents add to $n$. So you can apply FFT at the $m$th roots of unity where $m \geq 3n$ and is a power of 2. Then, cube each value pointwise to get values of $f(x)^3$. Then, apply inverse FFT, so $\frac{1}{m}$ FFT($f(x)^3$ values, $e^{-2\pi i/m}$) to recover the coefficients. Check if the coefficient of $x^n$ is 0. If it is, there are no 3 elements that sum to $n$, otherwise there is.

Runtime: $O(n \log n)$